Learning from Planned Data to Improve Pick-and-Place Efficiency

Liang Qin[1], Weiwei Wan[1], Jun Takahashi[2], Ryo Negishi[2], Masaki Matsushita[2], Kensuke Harada[1]

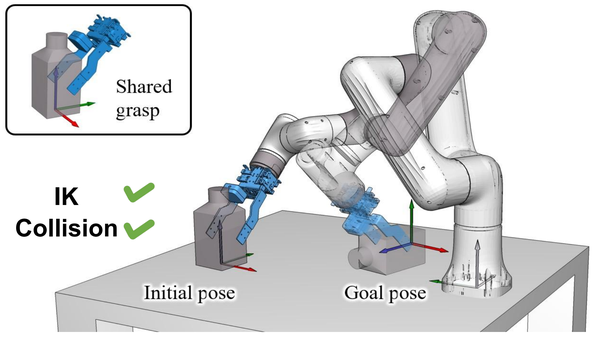
[1]The University of Osaka. [2]H.U. Group Research Inst. G. K., Japan.

Video Paper

Motivation: Shared grasps prediction

- Pick-and-place requires a grasp feasible at both the initial and goal poses.
- However, a grasp valid for picking may fail at placing due to IK or collision constraints. We call such a dual-feasible object-local candidate a shared grasp.

Shared Grasp Definition (Init & Goal)



Challenge: Exhaustive checking is expensive

- Analytical search checks IK and collision constraints for every candidate at both poses, which becomes expensive as the grasp set grows.

Method: Compositional Energy-based model(EBM) for shared-grasp prediction.

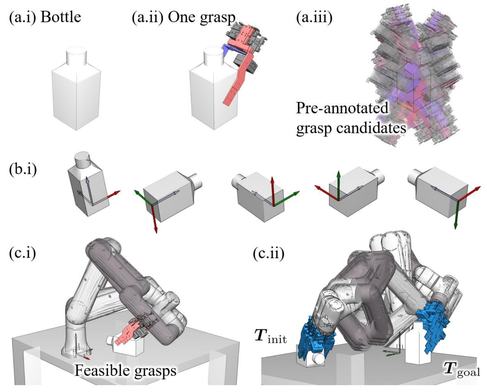
- **Step1: Modeling feasible grasp distribution by EBM.**

$$p_\phi(x) = \frac{\exp(-E_\phi(x)/t)}{Z_\phi} \quad Z_\phi = \int \exp(-E_\phi(x)/t)dx.$$

- Feasible grasp EBM purpose - Binary classification.
  - positive data energy < negative data energy.

$$E_{\phi_f}(\boldsymbol{T}, \boldsymbol{g}, w)$$

- Feasible grasp data collection
  - positive: $\mathbf{T}, \boldsymbol{g}, w$ pass ik / collsion check.



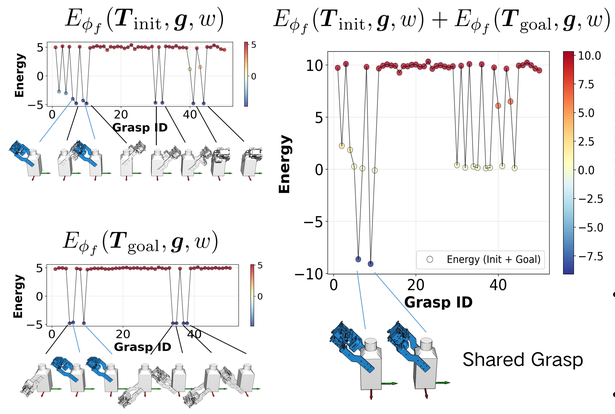
- **Step2: Compositional feasible grasp EBM to approximate the shared grasp distribution.**

$$p(x_1, x_2, ..., x_n) = \prod_i p_\phi(x_i) \propto \exp\left(-\sum_i E_\phi(x_i)\right)$$

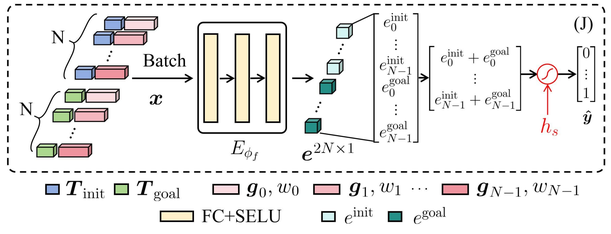
$$p(\boldsymbol{T}_{\text{init}}, \boldsymbol{T}_{\text{goal}}, \boldsymbol{g}, w) \propto \exp\left(-\left[E_{\phi_f}(\boldsymbol{T}_{\text{init}}, \boldsymbol{g}, w) + E_{\phi_f}(\boldsymbol{T}_{\text{goal}}, \boldsymbol{g}, w)\right]\right).$$

- Shared grasps are selected by thresholding summed energies, with batched inference for fast filtering.

$$\boldsymbol{\mathcal{G}}_{\text{shared}} = \left\{(\boldsymbol{g}, w) \mid E_{\phi_f}(\boldsymbol{T}_{\text{init}}, \boldsymbol{g}, w) + E_{\phi_f}(\boldsymbol{T}_{\text{goal}}, \boldsymbol{g}, w) < h_s\right\}$$



Proposed Method - Joint estimation (J)



- The inputs are processed in a batch-wise manner, enabling the EBM to compute all the energy values simultaneously in a single forward pass.
- The thershold caculated by max f1 socre of test dataset.

Experimental results & Baseline methods

Exp1: Training data efficiency

	100%			50%			15%			5%		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
J	94.0	95.3	94.6	94.2	91.3	93.0	90.0	81.9	85.7	85.9	77.8	81.6
D	92.9	95.2	94.1	91.2	94.4	92.8	78.9	81.5	80.2	70.1	80.4	74.9
L	90.7	97.3	93.8	89.7	96.5	93.0	78.5	93.2	85.2	77.4	88.5	82.6
F	98.2	98.7	98.4	98.0	98.4	98.2	94.5	96.6	95.5	92.1	94.0	93.0

Note F – Performance of the EBM model trained only for feasible grasp prediction, without incorporating shared grasp considerations.

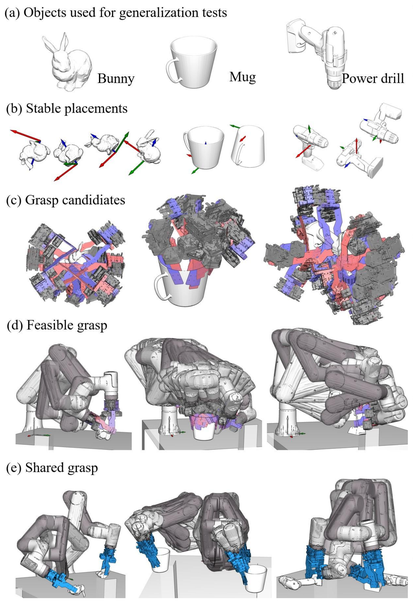
- Higher training data efficiency.
- Training data 15%(J) ≈ 50%(D/L) .

Exp2: Generalized to unseen grasp

	Model - 1			Model - 2			Model - 3			Model - 4		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
J	72.7	62.7	67.3	76.8	45.5	57.2	81.1	71.2	75.8	89.2	86.3	87.8
D	65.1	45.5	53.6	67.9	39.5	50.0	72.8	54.6	62.4	76.2	66.1	70.7
L	65.5	62.6	64.0	74.8	45.1	56.3	76.3	67.7	71.7	81.0	74.2	77.4
F	97.1	98.3	97.7	97.9	98.2	98.1	97.7	98.3	98.1	97.8	98.3	98.2

Note 1 Model 1 ∼ 4 trained on 57, 83, 109, 352 grasp candidates;
Note 2 The best values for shared grasp prediction are highlighted in lime.

- Generlizad to unseen grasp pose better.



Exp3: Generalized unseen objects

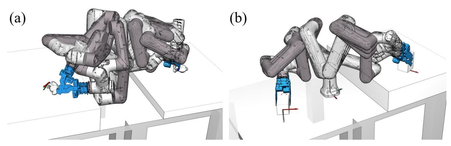
	Dataset	Bunny			Mug			Power drill		
		P	R	F1	P	R	F1	P	R	F1
J	Bt	1.6	100	3.2	53.3	62.1	57.3	23.6	91.9	37.6
	Bt+Bn	88.2	97.9	92.8	63.0	50.3	55.9	26.9	16.8	20.7
	Bt+Mg	1.7	80.5	3.4	80.0	97.7	86.7	20.8	87.7	33.7
	Bt+Pd	1.6	98.8	3.1	32.9	98.8	49.4	75.7	95.8	84.6
	Bt+Bn+Mg	80.6	98.8	88.8	76.6	96.6	85.5	29.6	47.4	36.4
	Bt+Mg+Pd	1.8	89.1	3.4	74.8	97.5	84.7	73.9	96.2	83.6
	Bt+Bn+Pd	85.8	97.2	91.1	47.1	54.5	50.5	74.2	95.1	83.4
D	Bt	4.2	0.5	0.1	14.7	6.7	9.2	21.5	69.0	32.8
	Bt+Bn	82.1	55.2	66.0	75.0	4.2	7.9	0.0	0.0	0.0
	Bt+Mg	0.0	0.0	0.0	78.6	85.7	82.0	0.0	0.0	0.0
	Bt+Pd	6.5	2.1	3.2	47.4	0.9	1.7	74.2	75.8	75.0
	Bt+Bn+Mg	80.6	42.0	55.2	76.0	82.8	79.3	0.0	0.0	0.0
	Bt+Mg+Pd	0.0	0.0	0.0	76.1	81.9	78.9	71.6	70.1	70.8
	Bt+Bn+Pd	85.3	52.7	65.1	36.9	0.8	1.6	71.3	73.9	72.5
L	Bt	6.2	7.6	6.8	84.4	39.6	53.9	45.5	17.7	25.5
	Bt+Bn	91.6	81.6	86.3	88.6	17.4	29.1	0.0	0.0	0.0
	Bt+Mg	5.6	0.6	1.2	80.6	92.3	86.0	0.0	0.0	0.0
	Bt+Pd	2.1	0.3	0.5	87.8	10.1	18.1	78.8	88.8	83.5
	Bt+Bn+Mg	88.9	72.0	79.5	79.6	91.7	85.2	0.0	0.0	0.0
	Bt+Mg+Pd	0.8	0.3	0.5	79.2	90.4	84.4	77.5	85.5	81.3
	Bt+Bn+Pd	90.3	76.6	82.9	72.7	6.7	12.3	78.0	87.6	82.6

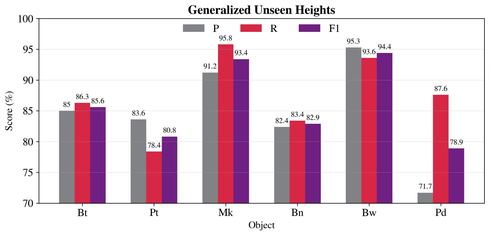
Note 1 Bt – Bottle; Bn – Bunny; Mg – Mug; Pd – Power drill;
Note 2 Gray: Results of seen objects; Black: Results of unseen objects.

- Generlizad to unseen / similar object shape.

Exp4: Generalized unseen heights

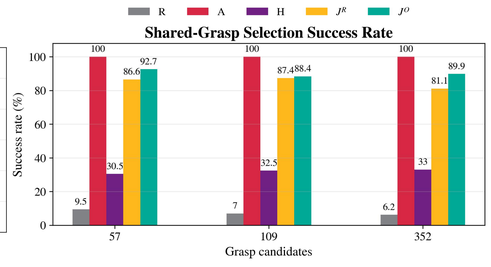
- Predict shared grasp on unseen heights.





Exp5: Motion planning success

- Compare to **R**andom, **A**nalytical, an **H**euristic(closed to init), and proposed method (**R**andom select / **O**pitimal energy select.)
- Shared grasp - necessary condition of planning.



Baseline methods

